AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 27 · STUDENT LESSON

Robustness across conditions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall Inspecting failures and anomalies. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of robustness across conditions. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of robustness across conditions.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Inspecting failures and anomalies. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Reproducible environment and records. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Predefining an analysis plan. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of robustness across conditions

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall Inspecting failures and anomalies. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

This week’s end goal

Explain robustness across conditions using the records and boundaries of this fictional investigation.

Notice the evidence

> Investigate the supplied Robustness across conditions case. Separate observations, given facts and assumptions before choosing an explanation.

Robustness checks examine a frozen procedure under declared varied conditions. Performance differences do not by themselves identify cause.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Robustness across conditions. Connect each labelled input, deciding step and result.

Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend **evidence** but cannot guarantee all conditions.

Robustness across conditions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Method passes9/10 on current sources and5/10 on outdated sources.	Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend evidence but cannot guarantee all conditions.	90% versus50%; gap40 percentage points. The outdated-source condition is associated with lower quality here; inspect mechanisms before general causal claims.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Robustness across conditions. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

Method passes9/10 on current sources and5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: Report only the combined 10/16=62.5%.

Predicted result

Prediction: The arithmetic is correct, but the condition gap becomes hidden.

Why the result follows

Reason: Combining records discards the slice distinction relevant to the deployment goal.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Robustness across conditions to explain a concrete decision, rather than only memorising definitions.

Word	Meaning
Robustness check	Evaluation under declared varied conditions
Condition-specific result	A score within one stated setting
Aggregate masking	A summary hiding differing condition outcomes

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Robustness across conditions, predict a step, then compare with its explanation.

Method passes9/10 on current sources and5/10 on outdated sources.

90% versus50%; gap40 percentage points. The outdated-source condition is associated with lower quality here; inspect **mechanism**s before general causal claims.

Worked explanation · Robustness across conditions

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Robustness across conditions

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

The facts we will use

Supplied facts

Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/27

Robustness across conditions

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Name both slices

Each condition has eight labelled cases.

2. Compute rates

Indoor is 87.5%; outdoor is 37.5%.



3. Bound robustness

The large observed gap motivates investigation rather than claiming condition independence.

CHANGE ONE THING

Report only the combined 10/16=62.5%.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Name both slices	Each condition has eight labelled cases.
2. Compute rates	Indoor is 87.5%; outdoor is 37.5%.
3. Bound robustness	The large observed gap motivates investigation rather than claiming condition independence.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

What this example does not establish

Keep the stated limit

Eight cases per slice do not establish stable population rates or the reason for the difference.

Keep the stated limit

Read the labelled picture: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

Keep the stated limit

Change one thing: Report only the combined 10/16=62.5%. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Robustness across conditions claim and explain a repair using the supplied evidence.

Robustness is bounded by the tested variations and independent **evidence** policy.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Robustness across conditions with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Robustness across conditions”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Robustness across conditions. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

> Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Robustness across conditions. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

Method passes9/10 on current sources and5/10 on outdated sources. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	

Record	Your evidence
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Robustness across conditions. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: Robustness is bounded by the tested variations and independent **evidence** policy. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Explain which evidence supports your conclusion and which claim remains unproved.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled robustness across conditions **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of robustness across conditions.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My limited claim and a useful next check

STUDENT LESSON · 13 OF 18

Study a complete case

Revisit the complete Robustness across conditions case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Robustness across conditions — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Method passes9/10 on current sources and5/10 on outdated sources.	Keep baseline, metric and eligibility aligned across conditions. Report condition counts and failures separately; new conditions may extend evidence but cannot guarantee all conditions.	90% versus50%; gap40 percentage points. The outdated-source condition is associated with lower quality here; inspect mechanisms before general causal claims.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Baseline/candidate on conditionA:6/8, 7/8; conditionB:5/8, 4/8. Compare paired rates.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Baseline/candidate on conditionA:6/8, 7/8; conditionB:5/8, 4/8. Compare paired rates.

My independent procedure, result and limitation

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Robustness across conditions, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

By the end, I can	How I will show it
Explain the mechanism and terminology of robustness across conditions	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Robustness across conditions without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of robustness across conditions

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building

block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The large observed gap motivates investigation rather than claiming condition independence.

1. Name both slices

2. Compute rates

3. Bound robustness

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Bound robustness The large observed gap motivates investigation rather than claiming condition independence.

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Report only the combined 10/16=62.5%. Which conclusion is supported by the supplied facts?

1. The arithmetic is correct, but the condition gap becomes hidden.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The arithmetic is correct, but the condition gap becomes hidden. Combining records discards the slice distinction relevant to the deployment goal.

Repair a missing building block

Review Inspecting failures and anomalies (ai-7/26). Goal: Explain the **mechanism** and terminology of inspecting failures and anomalies. Short practice: Calculate mean and median for 2,2,2,10.

What does the timeout log establish, and why is deleting that inconvenient record incompatible with the stated plan?. Return to this week and explain what you changed.

Review Reproducible environment and records (ai-7/24). Goal: Explain the **mechanism** and terminology of reproducible environment and records. Short practice: Trace the recorded mean for 2,4,6 and its changed result for 2,4,9.

Which artefacts must accompany the **output** so another team can identify the exact run?. Return to this week and explain what you changed.

Review Predefining an analysis plan (ai-7/21). Goal: Explain the **mechanism** and terminology of predefining an analysis plan. Short practice: Apply the recorded exclusion rule.

Can two readable model failures be removed, and how must a necessary later plan change be disclosed?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

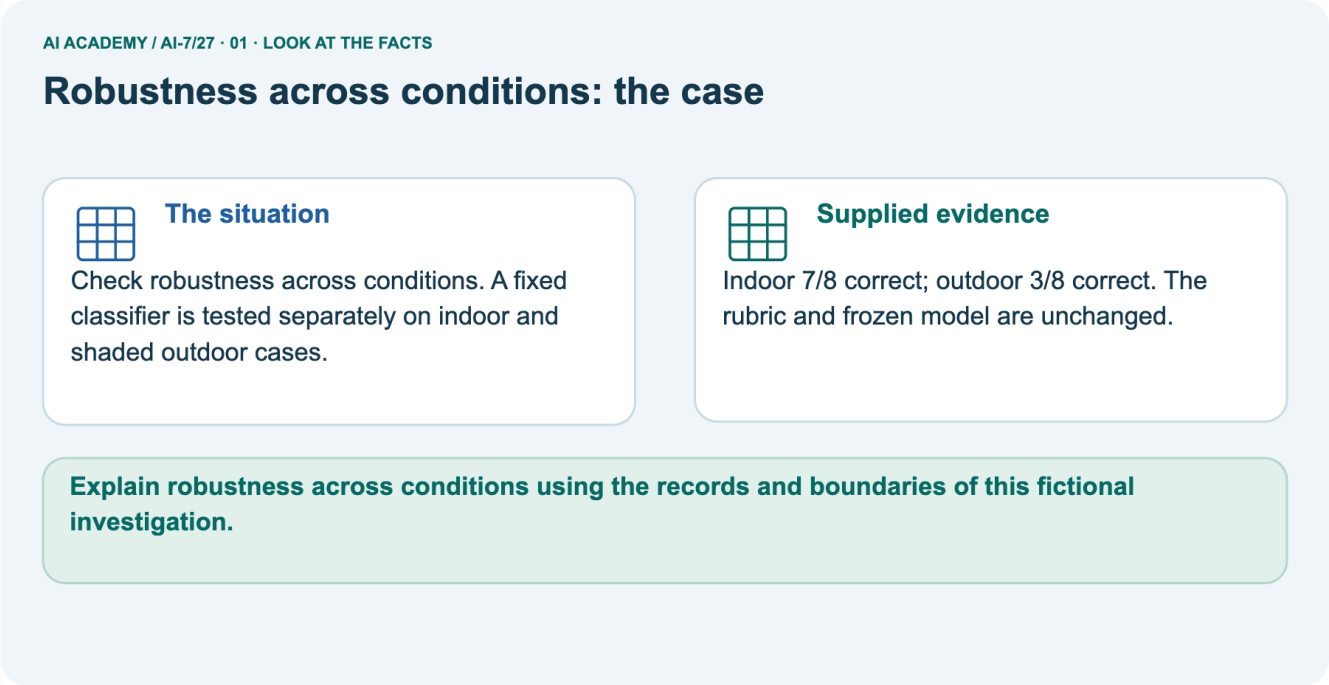
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Robustness across conditions graphic using its labels, deciding evidence and relevant limitation.

01. Robustness across conditions: the case

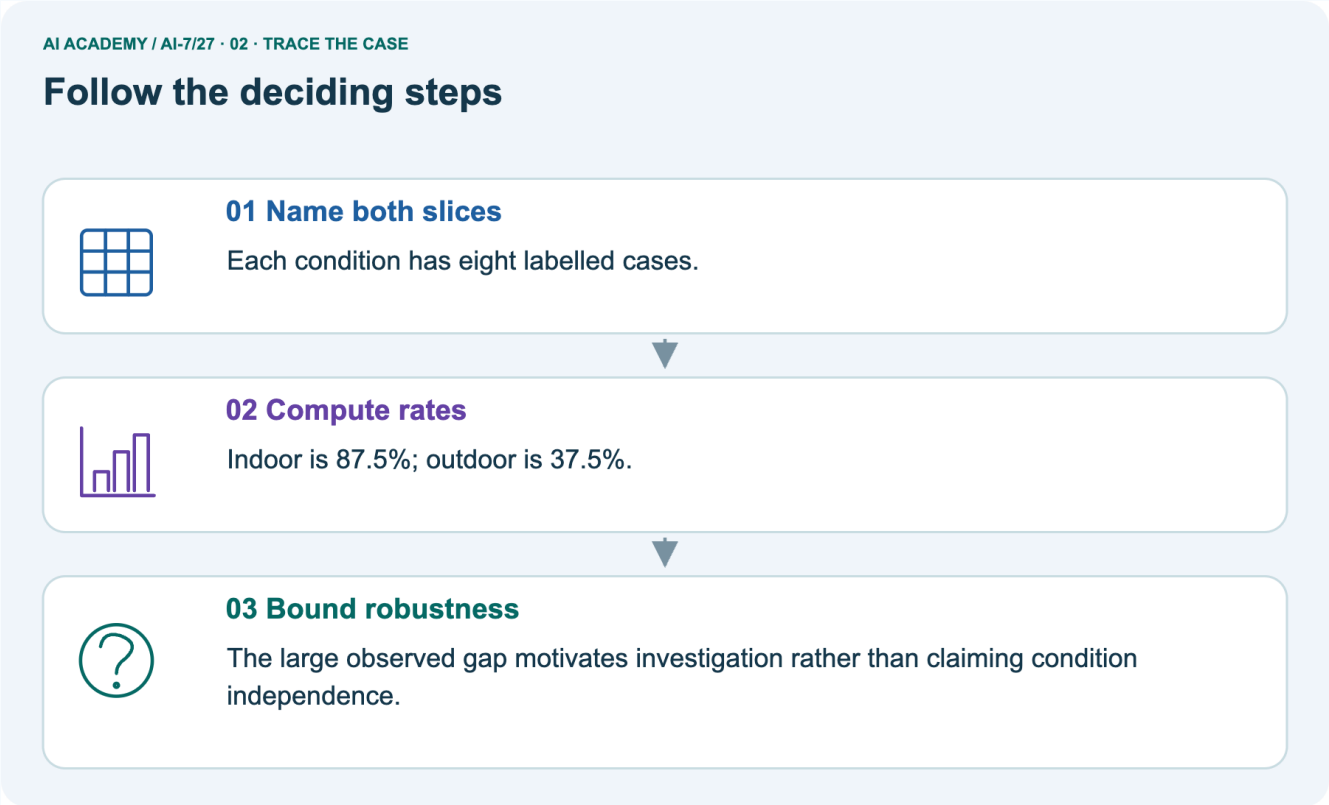


Follow the labelled picture with the matching explanation. Open the vector diagram

Check robustness across conditions. A fixed classifier is tested separately on indoor and shaded outdoor cases. Supplied facts: Indoor 7/8 correct; outdoor 3/8 correct. The rubric and frozen model are unchanged.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

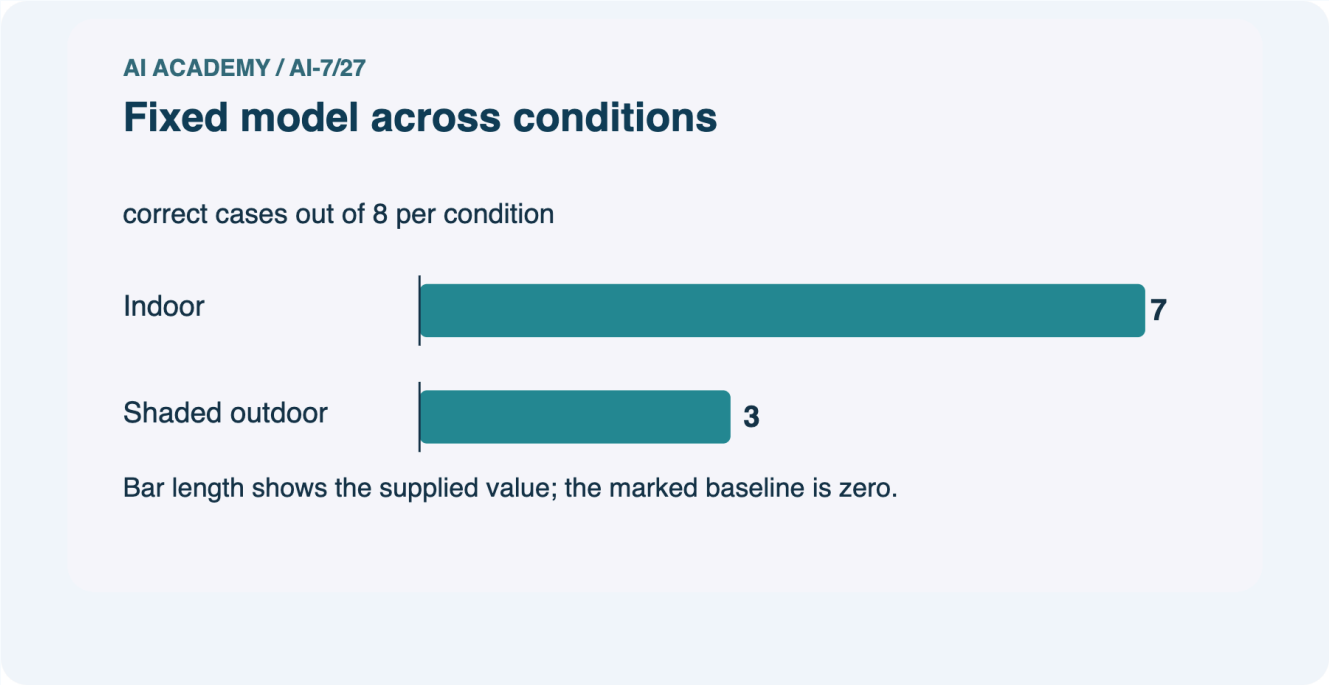


Follow the labelled picture with the matching explanation. Open the vector diagram

Name both slices: Each condition has eight labelled cases. → Compute rates: Indoor is 87.5%; outdoor is 37.5%. → Bound robustness: The large observed gap motivates investigation rather than claiming condition independence.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Fixed model across conditions

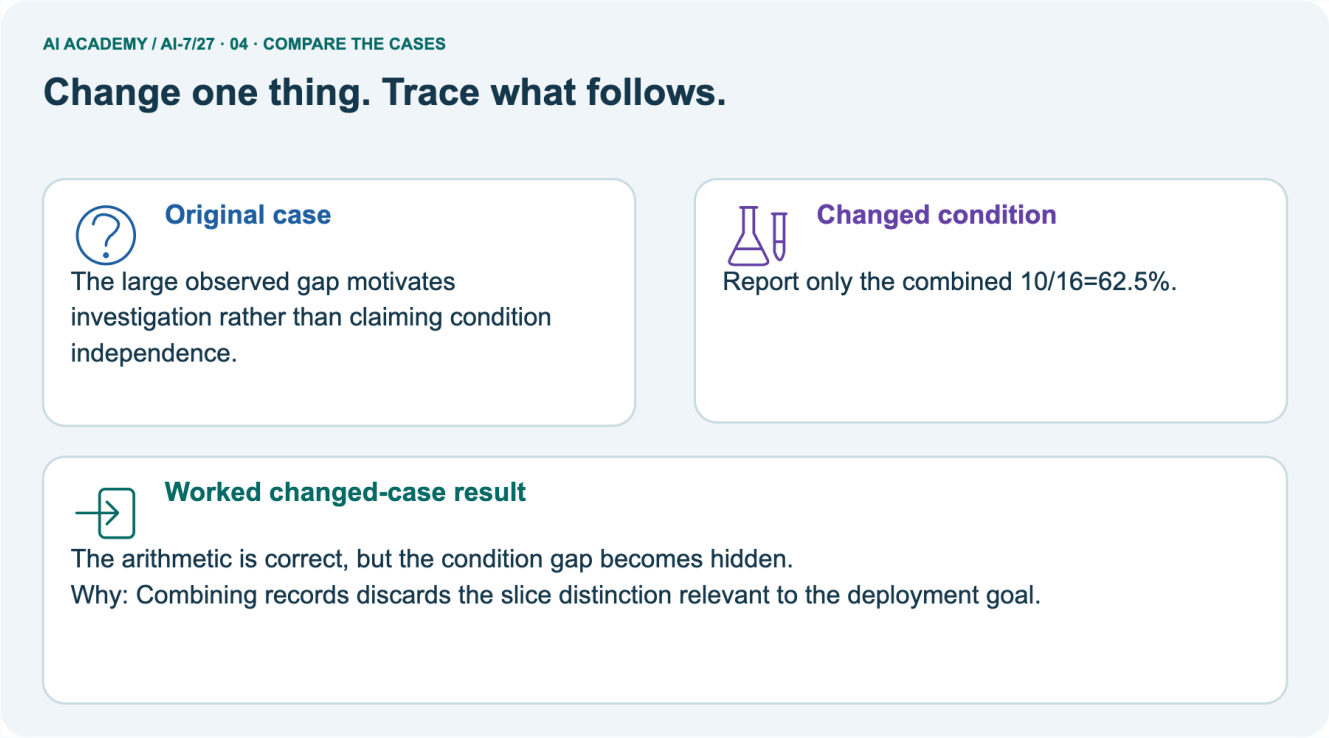


Follow the labelled picture with the matching explanation. Open the vector diagram

Fixed model across conditions. Units: correct cases out of 8 per condition. Indoor: 7. Shaded outdoor: 3.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



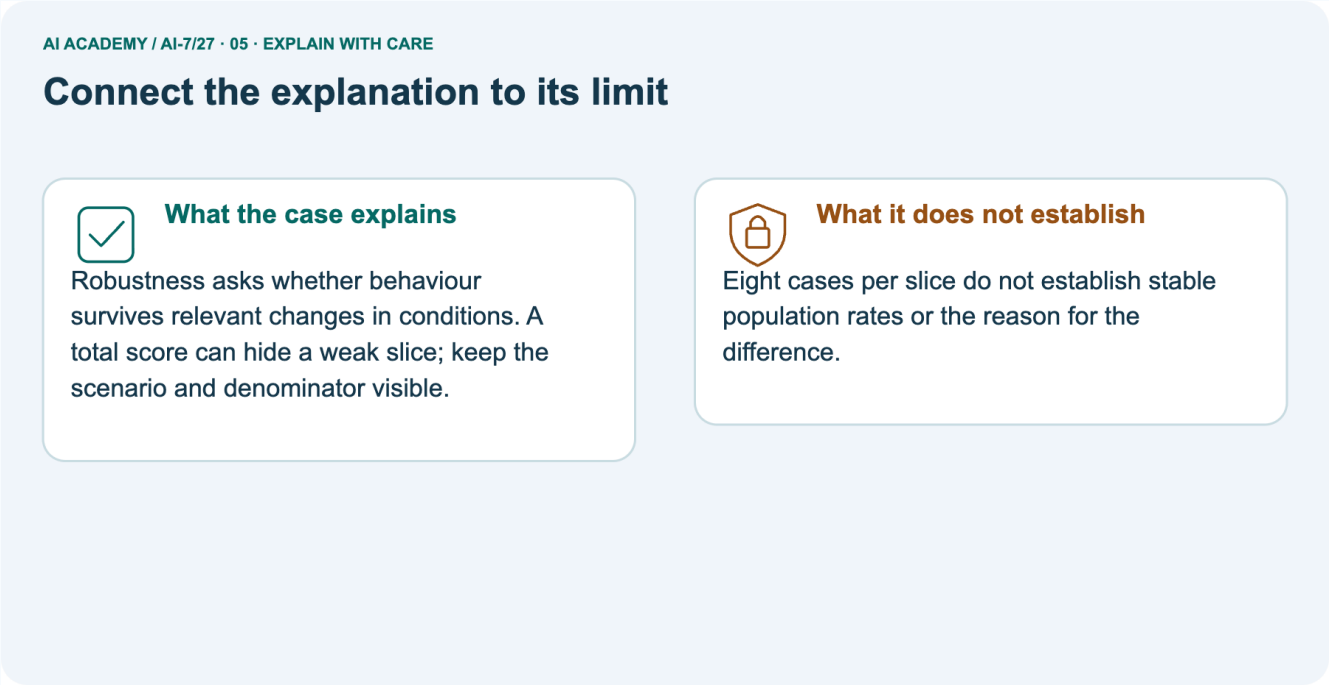
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The large observed gap motivates investigation rather than claiming condition independence. Changed condition: Report only the combined 10/16=62.5%.

Expected result: The arithmetic is correct, but the condition gap becomes hidden. Reason: Combining records discards the slice distinction relevant to the deployment goal.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



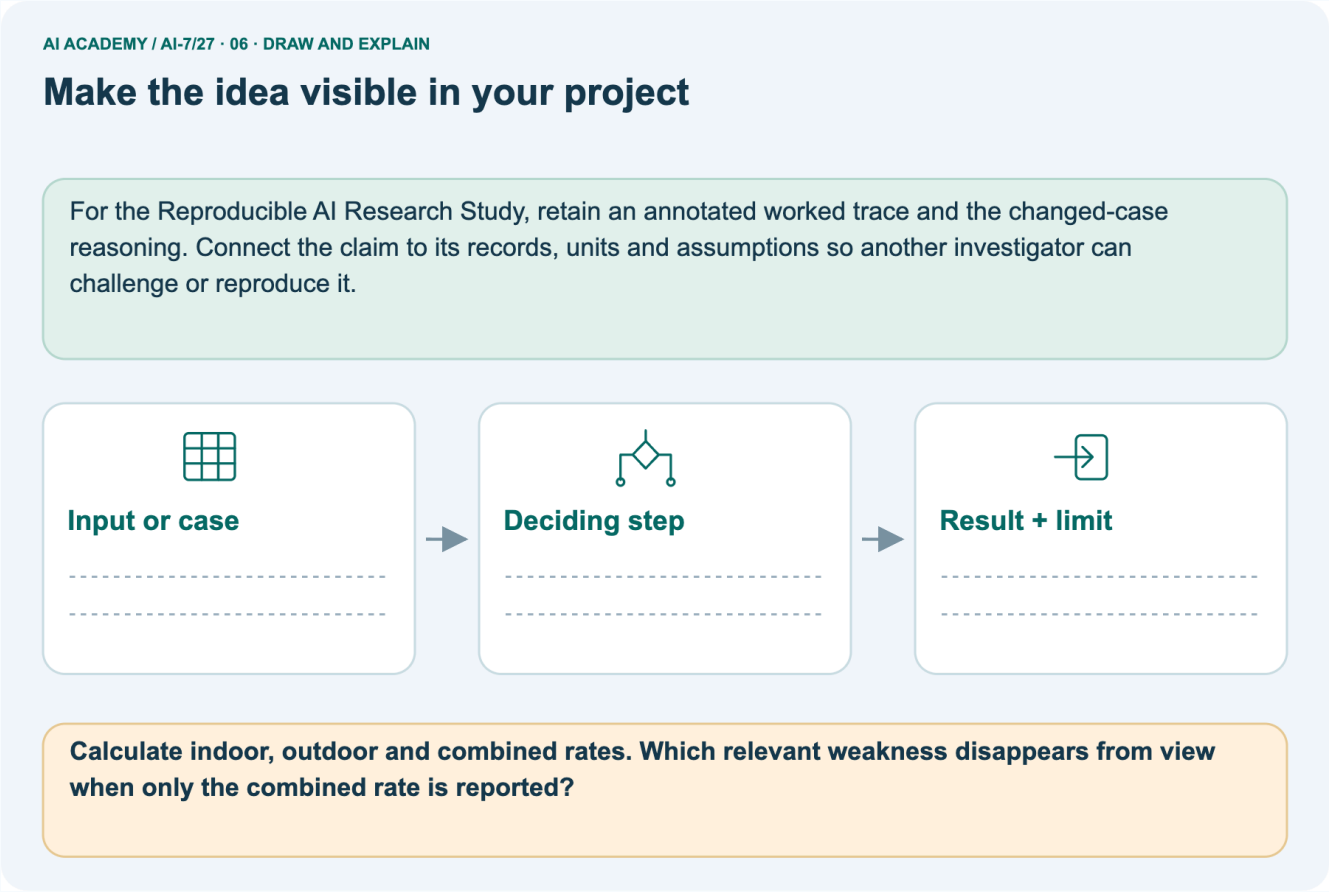
Follow the labelled picture with the matching explanation. Open the vector diagram

Robustness asks whether behaviour survives relevant changes in conditions. A total score can hide a weak slice; keep the scenario and denominator visible.

Limit: Eight cases per slice do not establish stable population rates or the reason for the difference.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate indoor, outdoor and combined rates. Which relevant weakness disappears from view when only the combined rate is reported?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.